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AI-Assisted peer review: a scoping review of governance, ethical-behavioral risks, and integrity

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ABSTRACT

Generative artificial intelligence (GAI) is reshaping peer review by reducing the cost of producing plausible, well-structured evaluative text while leaving the underlying cognitive work of reviewing only weakly observable. This article argues that AI-assisted peer review is, in practice, difficult to reverse because it is driven by convergent incentives—workload, speed, and competition—in a context where both reviewer effort and AI use are difficult to detect. Drawing on a qualitative analysis of recent literature and institutional policies, the article identifies three interrelated risks: effort outsourcing and accountability laundering, the limited detectability of AI-generated review reports, and adversarial vulnerabilities such as prompt injection embedded in manuscripts. In response, it proposes an auditable hybrid governance model centered on confidentiality, reviewer accountability, secure infrastructure, granular disclosure, and evidence-anchored auditing. The article concludes that preserving peer review as a practice of responsible judgment requires governance mechanisms that make AI-assisted workflows more verifiable without relying solely on bans or textual policing.

KEYWORDS

Peer review; generative AI; research integrity; accountability; governance

INTRODUCTION

Peer review is often described as an epistemic filter, but at a deeper level it operates as a social technology: a pact of trust that links competence, confidentiality, and accountability to editorial decisions and to the credibility of the scientific record (Committee on Publication Ethics, 2025; Nature Portfolio, n.d.). Within this arrangement, integrity is not sustained only by abstract norms; it depends on incentives, routines, and institutional mechanisms capable of reducing opportunities for deviation in low-observability environments, where the quality of review work is not directly verifiable by third parties. This intuition is clearly reflected in recent initiatives that incorporate integrity into the very design of evaluative processes, such as mandatory training in integrity and bias mitigation as a condition for serving as a reviewer and for gaining access to review materials (National Institutes of Health, 2023b, 2025).

The relevance of such measures increases once integrity is understood not merely as a matter of knowing the rules, but of creating institutional conditions under which ethical behavior becomes likely, traceable, and assessable. In this regard, recent analyses of ethics and integrity training highlight the limitations of metrics based on self-reports and recommend greater emphasis on behavioral outcomes and on how to measure real changes in practice rather than declared attitudes alone (Tammeleht & Lfstrm, 2025). This point is especially pertinent to peer review, whose value depends on demanding cognitive work that remains, to a large extent, invisible to editors and readers.

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Against this backdrop, generative artificial intelligence (GAI) enters peer review not only as a writing aid but as a technology that reorganizes the cost of cognitive effort. By making the production of plausible, well-structured prose cheaper, GAI facilitates the outsourcing of central evaluative steps, such as interpretive summarization, the identification of limitations, and the formulation of recommendations, precisely where diligence is least observable to editors and therefore most vulnerable to shortcuts. As a result, the risk shifts from occasional error to a quiet transformation in what counts as reviewing, in which the production of plausible prose tends to substitute for the patient verification that connects comment to evidence and critique to judgment, directly straining institutional requirements of confidentiality and reviewer accountability (Elsevier, 2025; Nature Portfolio, n.d.).

The situation becomes even more sensitive because use is already reported as widespread among reviewers, including in ways that contravene formal guidance, and because AI-generated reports may evade detection with relative ease, weakening strategies based solely on textual policing and strengthening incentives for performative compliance (Chawla, 2025; Naddaf, 2025). Recent studies reinforce this diagnosis by showing that, in AI conferences, the use of LLMs in review writing already appears at meaningful corpus-level rates and may influence evaluative outcomes themselves, tending toward higher scores and a greater likelihood of acceptance in borderline cases (Liang et al., 2024; Russo et al., 2025).

Moreover, a qualitatively new element reinforces the inadequacy of approaches centered only on prohibition and detection. Recent evidence shows that authors have embedded hidden instructions in manuscripts with the aim of influencing the AI tools used by reviewers, exploiting prompt-injection vulnerabilities (Gibney, 2025; Lin, 2025). This phenomenon shifts the threat model from the domain of mere fallibility to that of active manipulation: the manuscript ceases to be only an object of evaluation and becomes, potentially, an untrusted input to automated workflows, which strengthens the case for governance through infrastructure, sanitization, and explicit rules about what may or may not be processed by external systems.

Given the recent, heterogeneous, and multidimensional character of this literature – spread across empirical studies, analyses, editorials, commentaries, and institutional documents – a scoping review was chosen as the strategy for mapping the field. The review was guided by the following question: what evidence has been reported on governance approaches to the use of generative artificial intelligence in academic peer review, particularly with regard to confidentiality, human accountability, detectability, and risks of manipulation? On the basis of this question, the objective of the study was to map recent literature and relevant institutional documents, identifying convergences, tensions, and gaps in the contemporary debate. From this synthesis, the article discusses the elements of an auditable governance model that treats GAI as an instrument for support and triage rather than as a substitute for judgment, making the process more observable through secure infrastructure, specific transparency, and proportional auditing. In doing so, the article brings together, within a single normative and operational framework, three dimensions that are often treated separately: manuscript confidentiality, reviewer accountability, and adversarial vulnerabilities arising from AI-assisted workflows. The argument does not assume absolute irreversibility or institutional homogeneity; it advances the more modest claim that, under the incentives and constraints described in the recent literature, the diffusion of GAI is likely to be difficult to contain through abstract prohibitions alone, with variation across infrastructures, fields, and regulatory designs.

METHODS

This article adopts a scoping review design, reported in accordance with the PRISMA extension for Scoping Reviews (PRISMA-ScR). This methodological option was considered appropriate because of the recent, heterogeneous, and multidimensional nature of the literature on the reviewer's role in peer

review under generative artificial intelligence, which is distributed across empirical studies, analyses, editorials, commentaries, editorial policies, and relevant institutional guidance. The methodological strategy was designed to answer the review question formulated in the Introduction by mapping the available evidence, its convergences, tensions, and gaps, in order to support the discussion of an auditable governance proposal.

Review question and PCC framework

The review was guided by the following question: what evidence has been reported on governance approaches to the use of generative artificial intelligence in academic peer review, particularly with regard to confidentiality, human accountability, detectability, and risks of manipulation?

The formulation of the question followed the PCC framework recommended by JBI for scoping reviews. In the present study, the P component (Population/sources of evidence) encompassed reviewers, editors, journals, editorial policies, and institutional documents related to the peer review process. The C component (Concept) corresponded to the governance of the use of generative artificial intelligence in peer review, with emphasis on reviewer responsibilities and limits, the quality and reliability of reports, confidentiality, accountability, detectability, adversarial manipulation, and institutional regulatory mechanisms. The C component (Context) comprised academic peer review in scientific publishing, considering different disciplinary areas and editorial environments.

Eligibility criteria

The review considered sources published between 2024 and 2026 that directly addressed at least one of the thematic axes linked to the review question: reviewer responsibilities and limits in relation to generative artificial intelligence; the effects of GAI on the quality, specificity, or reliability of reports; governance of the review process under conditions of low detectability; risks of manipulation, fraud, and adversarial vectors associated with AI-assisted workflows; and reviewer incentives, conflicts, and the sustainability of review labor. As a complement, institutional sources and editorial policies without an explicit date were included when indispensable for grounding duties of confidentiality and accountability in the review process.

Eligible materials included empirical studies, analyses, reviews, editorials, commentaries, editorial policies, and institutional documents, provided that they contributed substantively to the review question. Sources focused exclusively on the use of AI in academic writing, teaching, or authorship, without a direct relationship to peer review, as well as duplicate documents, were excluded.

Information sources and search strategy

Searches were conducted in Web of Science, Scopus, PubMed, and Google Scholar, complemented by manual searching in journals with active debate on peer review and research integrity, as well as reference tracking in core texts. The search strategies combined descriptors and English-language variants related to peer review, generative artificial intelligence, and issues of governance, integrity, and risk.

In Scopus and Web of Science, the following search strategy was used:

((("peer review" OR reviewer* OR referee*) AND ("generative AI" OR "artificial intelligence" OR "large language model*" OR LLM* OR "AI-assisted peer review") AND (accountab* OR confidential* OR "review integrity" OR detect* OR "prompt injection" OR "paper mill*"))

In PubMed, an adapted title/abstract search strategy was used, with the following terms:

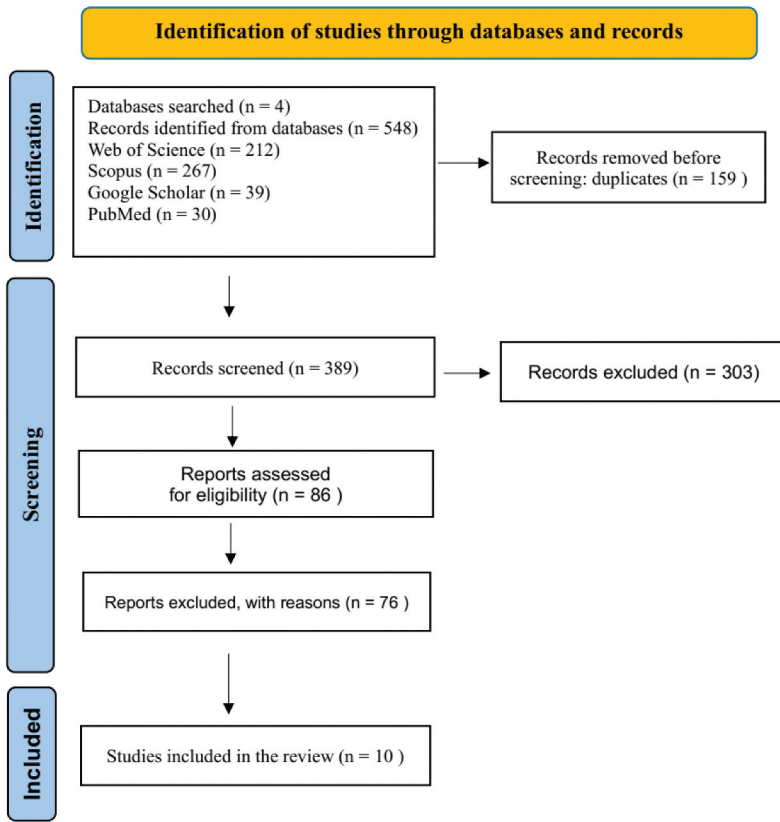


Figure 1. Flow diagram based on PRISMA 2020.

“peer review” OR reviewer* OR referee* AND “generative AI” OR “artificial intelligence” OR “large language model*” OR LLM OR LLMs OR “AI-assisted peer review” AND accountab* OR confidential OR “review integrity” OR detect* OR “prompt injection” OR “paper mill*”

In Google Scholar, because of the platform’s indexing characteristics and less standardized syntax, successive combinations of keywords were used rather than a single rigidly structured string, including the following terms: peer review, reviewer, referee, generative AI, large language model, LLM, AI-assisted peer review, accountability, confidentiality, review integrity, detection, prompt injection, and paper mills.

Selection and screening

Records identified in the databases were exported in bibliographic format and processed in RStudio, with the support of functions from the Bibliometrix package, for corpus organization and preliminary deduplication. After this stage, the remaining records were screened by title and abstract and, when potentially eligible, assessed in full text. Sources focused on the use of artificial intelligence in academic writing, teaching, or authorship, without a direct relation to peer review, as well as duplicate documents and texts that did not contribute substantively to the review question, were excluded.

Source selection was conducted in successive stages, as represented in the flow diagram based on the PRISMA 2020 model and adapted to the context of the present scoping review (Figure 1). First, the titles of the identified records were evaluated. Next, the abstracts of potentially relevant texts were read. Finally, the remaining sources were examined in greater depth, with attention to the abstract,

Table 1. Synthesis of the core corpus.

Author(s)	Year	Source focus	Finding/evidence	Main theme	Contribution
Li et al.	(2024)	Editorial policies	Restrictions on AI use based on confidentiality	Confidentiality	Grounds limits on external use
Wang and Gong	(2026)	Cross-disciplinary policies	Uneven regulation across fields and publishers	Governance	Supports regulatory heterogeneity
Tammeleht and Löfström	(2025)	Integrity measurement	Predominance of self-report and limited behavioral emphasis	Observability	Supports critique of fragile metrics
Barnett	(2025)	Citations and recommendations	Citations and self-citations influence reviews	Incentives and conflicts	Shows reputational distortions
Neveu and Neveu	(2025)	Fraud detection	Reviewers may detect fraud better than editors	Reviewer role	Reinforces epistemic centrality
Lin	(2025)	Prompt injection	The manuscript may function as an untrusted input	Adversariality	Supports the need for sanitization
Hoyt, Limon, and Chang	(2025)	Limits of AI in peer review	AI may support, but not replace, judgment	Support vs. replacement	Reinforces human oversight
Perlis et al.	(2025)	Limits of AI use	Human responsibility must remain central	Accountability	Supports nondelegable responsibility
Bauchner and Rivara	(2024)	Possible uses of AI	Legitimate use only in delimited tasks	Restricted use	Supports assistive deployment
Zou	(2024)	Responsible AI use	Ongoing adoption requires caution and guidance	Responsible use	Frames the contemporary debate

introduction, and objectives, in order to verify their fit with the review question and the established eligibility criteria. Screening was performed independently by two reviewers, and in cases of disagreement or impasse regarding inclusion, a third reviewer was consulted to reach a consensus decision. At the end of the process, the sources included were those that composed the analytical corpus of the review.

Data extraction, coding, and synthesis

The included sources were subjected to qualitative extraction and coding on the basis of analytic categories defined a priori and refined iteratively during reading. Among the initially established categories were: low observability of review work; low detectability of artificial intelligence use in reports; effort outsourcing and loss of specificity; reviewer accountability; manuscript confidentiality; adversariality and manipulation of evaluation workflows; reviewer incentives and conflicts; and available governance mechanisms, including infrastructure, disclosure, auditing, sanctions, and training. Table 1 summarizes the included sources according to their focus, the evidence mobilized, the main thematic dimension, and their analytical contribution to the review synthesis.

For purposes of traceability, a distinction was made between the review's core analytical corpus, composed of the sources that met the eligibility criteria and were counted in the flow diagram and synthesized in Table 1, and supplementary contextual materials used in the Introduction and Discussion, such as institutional policies, guidance documents, and additional references employed to frame the problem and discuss implications. This distinction was adopted to preserve methodological transparency between the formal review corpus and the broader argumentative support mobilized in the article.

Analytical synthesis and review limitations

The synthesis produced in this review made it possible to organize the findings into two interrelated planes. The first was diagnostic in nature and focused on identifying the limitations of approaches

based exclusively on prohibition and detection of artificial intelligence use in peer review. The second was analytical and propositional, oriented toward the discussion of institutional mechanisms capable of preserving confidentiality and human accountability while simultaneously expanding the observability of the review process, with emphasis on secure infrastructure, specific transparency, and proportional evidence-oriented auditing.

Because this study was based on published literature and public institutional documents, it did not involve data collection with human participants and therefore did not require submission to a research ethics committee. The limitations of the review include the rapid evolution of the field, which may require periodic updates to the mapping, and the heterogeneity of the included sources, which comprised both academic studies and editorial policies and institutional documents that differ in nature and evidentiary function.

RESULTS

General characteristics of the included sources

The review's core analytical corpus comprised 10 eligible sources, synthesized in Table 1. Additional institutional materials and editorial policies were used contextually in the Introduction and Discussion without being included in the formal corpus count. Taken together, the included sources evidenced a recent reorganization of the debate on peer review under generative artificial intelligence. The mapped literature concentrated on five recurring axes: human reviewer accountability, low detectability of AI use, confidentiality and the heterogeneity of editorial policies, adversarial risks associated with automated manuscript processing, and the sustainability of review labor. Although the sources varied in publication type and evidentiary function, they converged in recognizing that the incorporation of generative artificial intelligence shifts the focus from the report as a textual product to the review process as a practice of judgment and governance.

Human accountability and the redefinition of the reviewer's role

The included literature indicated a shift in the reviewer's role from a merely technical figure to an agent who is simultaneously epistemic and moral, responsible for sustaining judgment under conditions of low observability. In texts with both academic circulation and normative influence, the introduction of generative artificial intelligence into peer review was not treated as a simple gain in efficiency, but as a potential factor in the reduction of the reviewer's critical work by making the production of plausible, well-articulated reports cheaper. In this sense, the reviewer's role came to be described less as text production and more as an instance of accountability, that is, as the figure who answers for inferences, relevance, the sufficiency of the reasons advanced, and the prudential choices that sustain the review process, especially because generative systems may produce plausible critiques that are nevertheless generic or only superficially grounded, which reinforces the centrality of human accountability in the evaluative process (Maturo et al., 2025; Perlis et al., 2025).

Limited detectability and the shift of the problem to the process

Another recurring axis in the analyzed sources was the mismatch between formal rules and everyday practice. The mapped literature indicated that the use of AI by reviewers has become widespread, including in ways that conflict with editorial guidance, while the detection of AI-generated reports remains fragile. As a result, the problem no longer resides only in the final text of the report but comes to bear on the evaluation process itself. The included studies therefore suggest that more effective governance mechanisms depend on greater observability, traceability, and verifiability of judgment, without depending exclusively on unreliable textual detectors (Chawla, 2025; Naddaf, 2025; Russo et al., 2025).

Confidentiality and the heterogeneity of editorial policies

The review also identified, as a recurring finding, the centrality of confidentiality in institutional responses to the use of AI in peer review. The empirical literature on editorial policies showed wide variation between limited-permission regimes and explicit prohibition regimes. In particular, a substantial share of the policies analyzed severely restricts the processing of manuscripts by external tools, with justifications centered on confidentiality and risks of bias. Taken together, these findings indicate that peer review has come to be treated as a privileged locus of governance, especially because a substantial portion of the negative externalities of generative artificial intelligence emerges when the manuscript is processed outside controlled environments (Li et al., 2024; Wang & Gong, 2026).

Security, prompt injection, and manuscripts as untrusted inputs

One of the most sensitive domains identified in the corpus was the emergence of a security dimension in peer review. On the one hand, the included sources indicated that reviewers may be better positioned than editors to detect certain signs of fraud and inconsistency, reinforcing their role as integrity sensors. On the other hand, recent literature has documented the occurrence of prompt injection in manuscripts, with hidden instructions designed to influence the AI tools used by reviewers. These findings suggest that the manuscript ceases to be merely a passive object of evaluation and comes to be understood as a potentially untrusted input to automated workflows. In practical terms, this reinforces the need for stronger human validation, restrictions on the use of external tools, and a preference for controlled environments whenever automated assistance is employed (Lin, 2025; Neveu & Neveu, 2025).

Incentives, conflicts, and the sustainability of review labor

The mapped literature also refined understanding of the incentives and conflicts that cut across review work. Sources based on open-review data indicated mechanisms through which reputational interests may contaminate recommendations, even under greater transparency. At the same time, the corpus showed that the contemporary debate is not limited to risk and regulation, but also involves the sustainability of the reviewer's role in an ecosystem under pressure from volume and deadlines. In this context, part of the literature recognizes that generative artificial intelligence may have legitimate uses in triage and standardization, provided that this does not substitute for human judgment or reduce the rigor of decisions. Taken together, the findings reinforce the idea that the reviewer remains a scarce and critical resource whose responsibility cannot be dissolved through the automation of textual tasks (Barnett, 2025; Bauchner & Rivara, 2024; Hoyt et al., 2025; Zou, 2024).

Synthesis of review findings

Taken together, the included sources converged around a central point: the contemporary challenge is not to automate peer review, but to preserve the reviewer's role as an instance of responsible judgment in an environment marked by the diffusion of generative artificial intelligence, the limited detectability of its use, and the emergence of adversarial vectors associated with effort outsourcing. The review findings thus indicate a governance gap situated at the intersection of manuscript confidentiality, human accountability, and the design of institutional mechanisms capable of expanding the observability of the review process. This gap grounds the subsequent discussion of the elements of an auditable governance model.

DISCUSSION

The findings of this scoping review indicate that the incorporation of generative artificial intelligence into peer review does not merely introduce a new technical instrument, but reconfigures the institutional conditions under which review judgment is produced, observed, and governed. On the basis of these findings, this section discusses the analytical implications of that shift.

Low observability, effort outsourcing, and accountability laundering

Peer review operates under a structural asymmetry between what is normatively required and what is empirically observable. Reviewers are expected to read attentively, conduct critical examination, verify coherence, and exercise inferential prudence, but the system primarily observes the final artifact: a report delivered within a deadline. This low observability of cognitive work is constitutive of the model and has always represented a point of fragility. With the introduction of generative artificial intelligence (GAI), however, this feature ceases to be merely a limitation and becomes an active risk vector, because the technology drastically lowers the cost of producing a plausible, formally organized report without ensuring a proportional investment in diligence and verification.

In this context, effort outsourcing takes on a systemic character. What could once occur only marginally, as in rushed or excessively generic reviews, becomes more likely insofar as GAI allows central stages of the review process to be shifted to the tool, such as interpretive summarization, the enumeration of limitations, the formulation of recommendations, and even the drafting of the report itself. The critical point is not that such uses are always illegitimate, but that they alter the balance between judgment and text. The ethical boundary emerges when the tool ceases to function as peripheral support and instead occupies the core of judgment, producing an evaluation that is formally convincing but less anchored in the manuscript. It is in this sense that contemporary editorial policies and institutional guidance insist that the reviewer must remain responsible for the content of the report and that critical and original assessment cannot be delegated to generative systems (Elsevier, 2025; National Institutes of Health, 2023a; Nature Portfolio, n.d.; Perlis et al., 2025).

This displacement produces a particular form of institutional failure, here termed accountability laundering. This refers to a situation in which the human signature continues to function as a seal of legitimacy, while the link between critical statements and the actual verification performed by the signer is weakened. In practical terms, a report may retain an appearance of rigor by invoking “methodological points” and “typical limitations,” while losing specificity and fit to the evaluated manuscript. In normative terms, the damage is deeper: the system comes to reward fluency and rhetorical completeness rather than the verifiable tracing of reasons. The central risk, therefore, is not only epistemic but moral and institutional, because it shifts the basis of peer review from responsible judgment to the efficient production of plausible prose, eroding the expectation of accountability on which trust depends.

Limited detectability and why text-based enforcement fails

The most intuitive regulatory response to the risks of improper use of generative artificial intelligence (GAI) in peer review would be to prohibit or restrict its use and then detect possible violations. The problem is that peer review does not provide the minimal conditions for this strategy to work reliably. On the one hand, there is a mismatch between norm and practice: recent evidence indicates that the use of AI by reviewers is already widespread, including in ways that contravene editorial guidance (Naddaf, 2025). On the other hand, the detection of AI-generated reports remains technically and operationally fragile, which reduces the deterrent power of policies based on identifying “AI text” (Chawla, 2025).

When enforcement depends on limited detection capacity, the system tends to produce an undesirable equilibrium: formal compliance at the expense of practical opacity. The norm remains

the official language, but actual conduct migrates to less visible zones, where editors and institutions have limited capacity to calibrate risks, induce improvement, and learn from practice itself. This not only reduces the effectiveness of prohibitions, but may amplify the very risks that motivated them, especially confidentiality risks, because clandestine use tends to favor external tools and unauditable workflows. In such a scenario, text-based enforcement shifts the problem to a terrain in which the system has less governance, less transparency, and a weaker capacity for institutional learning.

The central point, therefore, is that low detectability is not merely a technical detail, but a relevant constraint of the problem under the conditions currently described in the literature. This does not mean assuming a static environment: platform watermarking, closed review portals, use logs, forensic stylometry, and other forms of enforcement may expand detection and deterrence capacity in some contexts. The issue is that such countermeasures tend to be partial, unevenly distributed across journals, and susceptible to strategic adaptation, so that they alter the conditions of the problem without, by themselves, providing a sufficient basis for governance. For this reason, a strategy centered exclusively on textual policing becomes insufficient. What is required instead is process-based governance: mechanisms capable of increasing the observability of judgment and the traceability of use without depending on the claim to detect authorship at the level of linguistic style. In other words, if trust cannot be anchored in detection alone, it must also be anchored in institutional design through secure infrastructure, specific transparency, and proportional evidence-oriented auditing, in line with the focus of contemporary policies on confidentiality and accountability (Elsevier, 2025; Nature Portfolio, n.d.).

Adversarial threats and manuscripts as untrusted inputs

Discussion of generative artificial intelligence (GAI) in peer review is often conducted as if the main problem were fallibility, expressed in hallucinations, biases, and genericity. Although that dimension remains relevant, recent literature points to the recognition of a qualitatively distinct threat: adversariality. There is now documentation of practices in which authors insert hidden instructions into manuscripts with the aim of influencing reviewers who use AI tools, exploiting prompt-injection vulnerabilities (Gibney, 2025; Lin, 2025). In this scenario, the manuscript ceases to be merely a passive object of evaluation and becomes a potential vector of attack on the judgment mechanism itself.

The implication for the reviewer's role and for governance is immediate. When a reviewer uses GAI to process manuscript text, the system may inadvertently incorporate instructions that would neither be relevant nor even visible to a human reader, but that are interpretable by the model. This alters the threat model of peer review: review no longer involves only content analysis, but also requires procedural hygiene and workflow security. In practical terms, it becomes inappropriate to treat the manuscript as a trusted input in automated workflows. In normative terms, this reinforces the centrality of controlled environments, sanitization, and clear restrictions on what may be submitted to external systems.

The adversarial dimension also deepens the inadequacy of strategies based solely on general prohibition. Even when the reviewer acts in good faith, using GAI on the raw text of the manuscript creates an attack surface that can be exploited by malicious actors, including in contexts where ex post detection remains weak. The boundary of the problem, therefore, is not limited to avoiding error; it extends to designing resilience in the review process against manipulation. In this shift, the reviewer is repositioned not only as an assessor of merit, but also as an element of system security.

Diagnostic synthesis: a threat-and-risk model with governance implications

Taken together, the discussion converges on a tripartite model of threat and risk associated with the use of generative artificial intelligence (GAI) in peer review. The first axis is behavioral-institutional: the low observability of review work favors effort outsourcing and accountability laundering, shifting review from judgment to the production of plausible prose.

The second axis is one of governance: low detectability weakens text-based enforcement, encourages clandestinity, and reduces institutional transparency and the capacity to calibrate practices. The third axis is adversarial: manuscripts may exploit vulnerabilities such as prompt injection and manipulate automated workflows, which requires submitted text to be treated as an untrusted input whenever it is processed by GAI.

The central implication of this diagnosis is that the problem is unlikely to be satisfactorily resolved through abstract prohibition and isolated stylistic detection. What is required instead is governance by process and infrastructure, oriented toward preserving confidentiality and accountability while increasing the observability of judgment. This implies explicit delimitation of acceptable uses, a preference for controlled environments, specific transparency regarding the use of GAI, and proportional auditing with reports anchored in manuscript evidence. This architecture does not eliminate tensions, but it responds to the central point identified in the discussion: when observability and detectability are limited, the integrity of the process depends less on presumptions of compliance than on institutional mechanisms capable of making practice verifiable.

Governance principles: confidentiality, accountability, and verifiability

The diagnosis presented here indicates that AI-assisted peer review should be treated as a problem of institutional governance rather than merely as a dilemma of individual conduct. The normative response should rest on principles compatible with low observability of review effort, low detectability of AI use in the final text, and the risk of adversarial manipulation. Three principles stand out as design invariants: confidentiality, accountability, and verifiability.

Confidentiality is a constitutive requirement of peer review and the condition that makes the process viable and legitimate. Submitted manuscripts contain unpublished ideas, potentially sensitive data, and, in many fields, direct intellectual-property implications. Contemporary policies converge in prohibiting reviewers from inserting manuscripts, or identifiable excerpts of them, into uncontrolled external tools, precisely to avoid exposure, undue retention, and circulation outside the review process. From a normative standpoint, confidentiality is not merely an instrumental attribute; it is part of the pact that legitimizes evaluation itself.

Accountability lies at the normative core of peer review. The moral value of peer review rests on the clear attribution of responsibility: a competent person stands publicly, at least before the editorial office, behind the reasons that guide a decision. When GAI lowers the cost of producing text and facilitates effort outsourcing, accountability becomes an axis that must be protected, because it is what distinguishes judgment from mere drafting. The implication is direct: any use of GAI must be compatible with reviewer imputability, that is, with the expectation that the reviewer can answer for every substantive claim in the report, including when it was initially suggested by a tool.

Verifiability is the operational principle that makes the first two workable in an environment of low observability. If the system cannot directly observe cognitive effort, it must observe indirect evidence and procedural traces that allow proportional auditing without violating confidentiality. Verifiability here does not mean making the process public, but making judgment auditable at appropriate layers: anchoring the report in concrete elements of the manuscript, providing transparency about the type of assistance used and the conditions under which the text was processed, and ensuring institutional capacity to sample and evaluate report quality systematically.

These three principles function as design constraints: the proposed architecture must simultaneously prevent content leakage (confidentiality), preserve nondelegable human responsibility (accountability), and create mechanisms that make that responsibility verifiable under realistic conditions (verifiability).

Architecture of the proposed model: auditable hybrid review

On the basis of these principles, the article proposes an architecture termed Auditable Hybrid Review (AHR). The aim of AHR is not to eliminate the presence of GAI, but to govern it so that efficiency gains do not become integrity losses. AHR acknowledges that automated assistance may exist, but establishes where it may operate, under which safeguards, and with which mechanisms of responsibility and control.

(a) Secure infrastructure (condition of possibility)

The first layer of AHR is techno-institutional: uses of GAI that involve processing the manuscript should occur, preferably, in environments controlled by the journal, consortium, university, or provider, with robust guarantees of nonretention and nontraining on submitted content, in addition to access controls and minimal logging. In the absence of secure infrastructure, the default rule should be restrictive for any use that requires the insertion of manuscript excerpts into external tools. This is not abstract rigorism; it is a rational response to confidentiality risks and to the fact that clandestinity tends to displace uses toward open, unauditable tools.

(b) Granular disclosure (functional transparency)

The second layer is informational: AHR requires specific and functional disclosure, not symbolic disclosure. Rather than asking merely “Did you use AI?” the regime should record: (i) the task for which GAI was used (for example, translation, restructuring the reviewer’s own report text, technical checking, manuscript summarization, or critique suggestion), (ii) whether manuscript content was processed and under what conditions (controlled environment or external tool), and (iii) whether any automated transformation of the report occurred beyond language polishing. Granularity is central for two reasons: it allows risk calibration, distinguishing lower-risk uses from higher-risk ones, and it reduces the gap between norm and practice by creating a regulated path for uses that are likely to occur, thereby diminishing incentives for concealment.

(c) AI as signal, not verdict (epistemology of support)

The third layer is epistemic: GAI may generate signals, never verdicts. This means that automated outputs should be treated as hypotheses, prompts for attention, or auxiliary checklists that require explicit human validation before they become substantive criticism. This rule directly targets the mechanism of accountability laundering. The tool may suggest, but the reviewer must verify and answer. In practical terms, this can be operationalized by requiring that substantive critiques in the report be anchored in manuscript evidence and that recommendations derived from GAI be reformulated in language that makes the reasoning and its links to evidence explicit.

(d) Evidence-anchored auditing (verifiability without breaching confidentiality)

The fourth layer concerns quality control: proportional auditing and sampling by anchoring. Because detectors of “AI text” are fragile, AHR replaces a logic of detection with one of verification. Evidence-anchored auditing consists in assessing, in samples of reports, whether relevant critical comments (i) refer to concrete points in the manuscript (sections, figures, or tables), (ii) provide sufficient justification for strong recommendations, (iii) avoid recyclable generalities, and (iv) avoid factual claims or references that cannot be verified. The focus is not on “catching” AI use, but on reducing the probability of reports that appear rigorous while lacking evidentiary support. This type of auditing also has dissuasive and pedagogical functions: it signals that the

quality of judgment will be assessed according to verifiable criteria, thereby reducing incentives for effort outsourcing.

(e) Responses to prompt injection and manuscripts as untrusted inputs (adversarial resilience)

The fifth layer concerns security against manipulation: AHR assumes that the manuscript is potentially an untrusted input to any automated workflow. Operational responses include file and text sanitization before any GAI processing (removal of invisible text, normalization of formatting, and detection of suspicious elements), restrictions on the tools and environments in which the manuscript may be processed, and explicit instructions that discourage the use of external systems with the full manuscript text. In more robust regimes, journals may offer internal tools with adversarial filters and access logs for later investigation, thereby reducing the attack surface associated with effort outsourcing.

Taken together, these layers form a governable architecture. Infrastructure reduces confidentiality risk, disclosure creates functional transparency, the signal-not-verdict rule preserves human judgment, evidence-anchored auditing makes practice verifiable, and the adversarial module protects against manipulation. AHR therefore transforms a problem often treated as one of individual ethics into a problem of institutional design with implementable components.

Institutional implications and trade-offs: time, workload, equity, and costs

The adoption of a regime such as Auditable Hybrid Review (AHR) entails inevitable trade-offs. The model's virtue does not lie in promising the elimination of tensions, but in making them explicit and governable so that efficiency gains are not obtained at the expense of integrity.

With respect to time and editorial speed, pressure for rapid decisions is one of the forces driving GAI in review. AHR may, in some scenarios, reduce timelines by automating peripheral tasks and standardizing checks, but it may also introduce deliberate frictions, such as granular transparency and sample-based audits. These costs, however, should be compared with the institutional costs of low-quality reviews, which often take the form of rework, reevaluation, post-publication corrections, editorial conflict, and reputational erosion. In other words, part of the time “saved” through shortcuts tends to reappear as time “paid” in the form of integrity failures.

Regarding reviewer workload, there is a real risk that governance may increase bureaucracy and discourage reviewers. For this reason, AHR should be implemented with friction minimization: disclosures can be structured as checkboxes and short categories, audits can be infrequent and risk-oriented, and internal tools can reduce manual work. The logic is to shift effort from the reviewer to support systems and to reduce, rather than increase, peripheral cognitive load. Even so, the model assumes a decisive normative thesis: reducing workload cannot imply reducing responsibility. The system should relieve mechanical tasks, not outsource judgment.

At the level of justice and equity, GAI may offer distributive benefits, for example by helping non-native reviewers draft clearer reports and by reducing linguistic biases that penalize authors. At the same time, it may introduce systematic biases derived from training data and reinforce power asymmetries when used to produce reviews that are rhetorically strong yet epistemically weak. AHR makes it possible to capture low-exposure benefits, such as language editing of the reviewer's own report, while restricting higher-risk uses, such as manuscript summarization in external tools. In addition, evidence-anchored auditing promotes fairness by requiring that strong critiques be tied to manuscript evidence, thereby reducing arbitrariness and the circulation of generic opinions that tend to affect authors from less prestigious groups disproportionately.

There are also trade-offs related to costs and institutional capacity. Secure infrastructure and access records involve costs and may be unfeasible for smaller journals. For this reason, AHR should be conceived in levels. A minimal regime may consist of a clear prohibition on the external upload of manuscripts, mandatory granular transparency, and light evidence-anchored auditing. An

intermediate regime may add internal tools for triage and sanitization. An advanced regime may include private GAI environments and robust mechanisms for adversarial mitigation. The normative point is that governance costs should be proportional to risk and journal impact, but that there remains a minimum level of governance necessary when the aim is to preserve integrity in a context of low detectability.

AHR makes explicit a fundamental exchange: efficiency gains are acceptable only if they preserve confidentiality and accountability and if they are accompanied by mechanisms that make the process verifiable. This is the condition under which peer review can continue to sustain trust rather than merely accelerate decisions.

Research agenda and implications for editorial policy

The contemporary debate on GAI-assisted peer review tends to oscillate between two incomplete responses: abstract trust in reviewer self-regulation or formal prohibition supported by fragile detection mechanisms. The architecture proposed in this article assumes that governance must be evidence-based and that, accordingly, the relevant frontier is empirical: which institutional mechanisms reduce effort outsourcing and accountability laundering without undermining the sustainability of the system? A coherent research agenda must therefore test hypotheses about reviewer behavior, report quality, and the effectiveness of instruments such as granular disclosure and evidence-anchored auditing, while also delineating metrics capable of monitoring integrity without compromising confidentiality.

Testable hypotheses: reviewer behavior, report quality, and the effectiveness of interventions

A first family of hypotheses concerns the effect of GAI on diligence and cognitive effort. In behavioral terms, the central prediction is that access to GAI will reduce the time needed to produce a report and increase textual fluency, but may reduce anchoring and specificity when the tool is used to generate critical content rather than merely to support form. This can be tested experimentally through simulated review tasks and standardized manuscripts, randomizing conditions of access to GAI (no AI; AI for language revision of the reviewer's own report; AI for manuscript summarization; AI for critique generation) and measuring evidence density, critique accuracy, and the rate at which deliberately planted problems are detected. The expected result is a gradient: the more GAI is used in the core of judgment, the greater the risk of plausible but less verifiable reports.

A second family of hypotheses focuses on accountability laundering as a mechanism of conduct. The prediction is that, under time pressure, the presence of GAI increases the probability that the reviewer will accept unverified suggestions, especially when the final report displays fluency and rhetorical completeness. A relevant test is to measure whether reviewers with GAI become less likely to critically revisit their own claims, for example by accepting generalizations or references without checking them. A useful design is to introduce controlled "traps" (simple statistical errors, inconsistencies between text and table, retracted references) and observe detection rates and patterns of justification, distinguishing between anchored critiques (those referring to evidence in the manuscript) and unanchored critiques (generic ones). The hypothesis is that GAI expands the production of unanchored critiques when it is used to generate content, but not when it is used only to improve the language of text already produced by the reviewer.

A third family of hypotheses tests the effectiveness of granular disclosure. The normative assumption is that specific transparency reduces clandestinity and creates conditions for calibrating risk, but this must be demonstrated, because disclosure may also generate displacement effects, such as underreporting due to fear of sanctions or substitution toward less traceable tools. One testable hypothesis is that disclosure models with clear categories and low friction (for example, a short checklist) increase disclosure rates and reduce the use of external tools for manuscript processing

when compared with vague policies or prohibitions that provide no permitted-use pathways. Another hypothesis is that granular disclosure indirectly improves the observable quality of reports by signaling that the process is governed and auditable, thereby reducing incentives for effort outsourcing.

A fourth family of hypotheses evaluates the effectiveness of evidence-anchored auditing as a governance mechanism in low-detectability environments. The hypothesis is that sample-based audits, combined with clear criteria for anchoring critiques to manuscript evidence, reduce the proportion of generic reports and increase justification density, even when the use of GAI remains possible. This effect can be tested in the field by comparing editorial periods before and after the implementation of light auditing (for example, auditing 5% of reports and providing feedback to reviewers) and monitoring changes in metrics of specificity and correctness. A related and testable corollary is that evidence-anchored auditing has a pedagogical function, improving average quality by creating shared standards for what counts as a “good review.”

A fifth family of hypotheses investigates adversarial threat, which is qualitatively distinct from error and bias. The prediction is that prompt injection and other manipulation strategies will have greater impact when the reviewer uses GAI to process the raw text of the manuscript, especially in external tools without sanitization. This allows controlled tests: versions of the same manuscript with and without hidden instructions can be evaluated by reviewers under different conditions of GAI use, with susceptibility to adversarial instructions measured across conditions. The hypothesis is that sanitization and controlled environments reduce the effect, whereas the ad hoc use of external tools increases vulnerability. This program is relevant not only to the integrity of the process, but also to the technical justification of institutional restrictions regarding where and how manuscripts may be processed.

Recommended metrics for monitoring integrity without breaching confidentiality

Monitoring integrity in peer review is particularly delicate because the data are sensitive and confidentiality is a pillar of the process. The recommended metrics, therefore, should satisfy two criteria: they should be informative for governance and they should minimize content exposure. A promising strategy is to prioritize observable metrics of process and report quality, without any need to make manuscripts or reports public.

A first class consists of anchoring and verifiability metrics. Examples include the proportion of critical comments that cite specific sections, figures, or tables; the density of justifications per strong recommendation; the presence of clear operational recommendations (specific actions that the author can take); and an index of generality estimated from linguistic patterns or the repetition of generic structures, provided that such metrics are used only for triage and are complemented by human sampling. The focus is to measure whether the report demonstrates verification work, not whether it was “written by AI.”

A second class consists of coherence and accuracy metrics, which can be implemented through internal audits. Examples include the rate of “false positives” (critiques that do not apply to the manuscript), the detection of nonexistent or inappropriate references in the report, and the incidence of unsupported factual claims. These metrics can be generated in internal sample audits, with aggregated results reported without exposing content.

A third class consists of functional compliance metrics associated with granular disclosure and use governance. Examples include the disclosure rate of GAI use by task type (translation, editing, summarization, critique generation); the proportion of disclosures involving manuscript processing outside controlled environments; and the frequency of reviews flagged for violations of confidentiality policies. When reported, these metrics should be aggregated and nonidentifiable, reducing exposure risk and preserving trust between reviewers and editorial offices.

A fourth class consists of sustainability and workload metrics, which are essential to prevent governance from making the system unworkable. Indicators such as average review time, invitation-decline rates, time to editorial decision, and the distribution of review burden by area can indicate whether an intervention increased friction to the point of compromising recruitment. These metrics

make it possible to adjust the governance regime so as to minimize bureaucratic cost without abandoning core principles.

A fifth class consists of post-publication outcome metrics, which help evaluate the indirect effectiveness of the review process. Examples include rates of corrections, expressions of concern, and retractions; patterns in post-publication commentary; and the incidence of methodological problems identified after publication. These metrics are imperfect, but they may indicate whether changes in the review process correlate with improvement or deterioration in quality.

Alternative perspectives and governance complementarities

A balanced analysis of the problem requires consideration of alternatives that, although they do not eliminate the tensions described here, may form part of plausible institutional responses. Strict prohibition regimes with technical enforcement inside review portals may be appropriate in highly controlled environments, especially when the goal is to prevent external manuscript upload and to log interactions with authorized tools. Similarly, “human-only” review certifications may have symbolic and procedural utility for sensitive manuscripts or for specific stages of the process. The common limit of these strategies is that their effectiveness depends on verification capacity and on infrastructure that is unevenly distributed across journals; without these conditions, they risk displacing the problem toward clandestine or merely self-attested uses.

Other alternatives also offer partial gains. Open peer review and reputational systems may increase transparency, reputation, and accountability, but they do not by themselves solve the confidential processing of manuscripts in external tools or the vulnerability to prompt injection. Measures aimed at reducing demand, such as qualified desk rejection, editorial triage, and better workload distribution, address an important structural driver of effort outsourcing. On this reading, Auditable Hybrid Review does not necessarily replace these strategies; rather, it functions as an architecture compatible with different institutional combinations, capable of integrating restrictions, transparency, auditing, and technological support at levels proportional to the risk and capacity of each journal.

CONCLUSION

This scoping review began from the observation that peer review assisted by generative artificial intelligence (GAI) already operates as an important force of change in the scientific ecosystem and, under the conditions described in the recent literature, tends to be difficult to contain through formal prohibitions or textual policing alone. The mapped findings indicated that this transformation rests on three simultaneous conditions: the low observability of the reviewer’s actual work, the low detectability of AI use in reports, and the emergence of adversarial vectors that exploit effort outsourcing itself. Taken together, these elements suggest that GAI may shift the balance between judgment and text, increasing the risk of plausible but weakly anchored reports and favoring forms of accountability laundering in which the human signature preserves an appearance of imputability without ensuring proportional verification.

The question guiding the review was: what evidence has been reported on governance approaches to the use of GAI in academic peer review, particularly with regard to confidentiality, human accountability, detectability, and risks of manipulation? The answer developed here points to a shift from text-based enforcement toward governance by process and infrastructure. More specifically, preserving peer review as a practice of responsible judgment requires institutional arrangements capable of making the process more observable and verifiable without violating confidentiality, through combinations of secure infrastructure, specific transparency, proportional auditing, and safeguards against adversarial manipulation.

The article makes three contributions. The first is analytical: by integrating low observability, low detectability, and adversariality into a single threat model, the review makes explicit why approaches based solely on prohibition and detection are structurally fragile and may lead to opacity,

clandestinity, or merely performative compliance. The second is normative: by re-centering confidentiality and accountability as design invariants, it offers criteria for distinguishing legitimate support uses from delegations that erode the reviewer's responsibility. The third is operational: the proposal of Auditable Hybrid Review organizes implementable mechanisms – such as evidence-anchored auditing, granular disclosure, and governance by secure infrastructure – while also situating these mechanisms within an empirical agenda that is open to testing and institutional variation.

This work has limitations. It is a scoping review based on academic literature and institutional documents, rather than an original empirical study; for this reason, the propositions developed here require validation through experimental designs and field evaluations, especially with respect to the effects of disclosure and auditing on actual reviewer behavior. In addition, rapid technological change may alter the relationship among detectability, quality, and adversariality, requiring periodic updating of the proposed recommendations and metrics. There is also a relevant institutional trade-off: increasing verifiability may raise frictions and costs, with the risk of reducing reviewer participation if governance is implemented with excessive bureaucracy or without adequate technological support. In this sense, the proposed architecture should be understood as a layered arrangement, adjustable to the capacities and risk profiles of different journals.

Despite these limitations, the general implication remains clear. Even if detection and enforcement capacities evolve, the legitimacy of peer review will depend less on abstract declarations of principle and more on institutional mechanisms capable of transforming trust into verifiability. Preserving peer review as a practice of judgment requires preserving the reviewer as the locus of accountability and protecting the manuscript as a confidential object, while also recognizing the reality of adoption and the new vulnerabilities of the process. The challenge is not simply to accept or reject GAI, but to govern its presence in ways that prevent peer review from being reduced to the efficient production of plausible prose. On this view, the sustainable future of peer review is likely to be hybrid, but it must remain, by design, human and institutionally accountable.

AUTHOR CONTRIBUTIONS

CRedit: **Luiz Alberto Pilatti:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Project administration; **José Roberto Herrera Cantorani:** Validation, Formal analysis, Writing – review & editing; **Fabiana Fátima Do Prado Sedelak Pinheiro:** Validation, Writing – review & editing.

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